Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/16/2025
PI Name:	Weber, Richard Ph.D.		
Project Title:	The Origin of Fragility in High-Temperature Oxide Liquids		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rweber@matsdev.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	847-612-8597
Organization Name:	Materials Development Inc.		
PI Address 1:	3090 N Daniels Ct		
PI Address 2:			
PI Web Page:			
City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Japanese Space Agency (JAXA) AO for Electrostatic Levitation Furnace (ELF) experiments on ISS
Start Date:	08/15/2019	**End Date:**	09/30/2025
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sansoucie, Michael	**Contact Phone:**	256.544.5269
Contact Email:	michael.p.sansoucie@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NSSC information (Ed., 10/9/24) NOTE: End date changed to 9/30/2024 per NSSC information (Ed., 3/5/24) NOTE: End date changed to 9/30/2023 per NSSC information (Ed., 6/20/23) NOTE: End date changed to 9/30/2022 per M. Sansoucie/MSFC (Ed., 1/28/21) June 2019: Note project had preliminary work performed Feb-August 2019		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1288 ; 80NSSC19K0518		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is the United States (US) portion of an international Japan Aerospace Exploration Agency (JAXA) program to investigate the properties and structure of molten and supercooled metal oxides using the Electrostatic Levitation Furnace (ELF) on the International Space Station (ISS) and ground-based synchrotron x-ray and neutron studies. Four prototype compositions have been levitated and melted on ISS. Measurements of thermophysical properties, density, thermal expansion coefficient, viscosity and surface tension were made over a wide temperature range. Samples, including glasses, were recovered for ground based analysis and comparison with materials made on Earth. The work underpins development of a materials property-structure data set that can help in development of new materials and in meeting requirements of material utilization and processing in space. Additional samples are being prepared for measurements on ISS in 2023. Work is being published to provide data to the broader scientific community. The project team brings together the skills and experience needed to implement a complex research program. Reference: [1] S. Kohara, et al., “The origin of fragility in high-temperature oxide liquids - towards fabrication of novel non-equilibrium oxides,” JAXA flight project funded in 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Developing an advanced understanding of the properties and structure of liquids that can be transformed into glasses is a high value enterprise that benefits the $300B global glass materials industry. Specific advances expected from this research will impact development of functional glasses by providing high accuracy thermophysical property and atomic structure data that can be used to optimize models used to design new materials from first principles. The titanate liquids being studied have high value applications. The compositions also relate to lunar mare regolith that contains relatively large amounts of titanium oxide. The work has led to discovery a new type of glass with highly coordinated titanium ions. Work is/has been published and a US patent on an optical device glass was filed.
Task Progress:	The project was very successful. Approximately 30 samples covering seven chemical compositions were run on the Japanese Space Agency's electrostatic levitation furnace (ELF) on the International Space Station (ISS) and returned to Materials Development Inc. (MDI) laboratories for analysis. The overall success rate for levitation was ~65%. Property measurements were made on all samples that were melted. Some key accomplishments were to: • Determine the structure of a non-conventional glass forming system based on a titanium dioxide network forming component. • Measure the density, thermal expansion coefficient, surface tension and viscosity of normal and supercooled liquids. • Investigate the properties of several glasses made in microgravity. • Measure the optical properties of rare earth doped glasses. • Publish and present results prior to uploading them to the NASA Physical Sciences Informatics (PSI) database. • Upload data from this work to the PSI system. • File two patents on glass materials made in reduced gravity. These are being pursued in commercial activities.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Benmore CJ, Wilke SK, Lipke D, Weber R. "High temperature diffraction from aerodynamically levitated materials." Advances in Physics: X. 2026 Mar 14;11(1):2642120. https://doi.org/10.1080/23746149.2026.2642120 , Mar-2026
Articles in Peer-reviewed Journals	Tolliver J, Neumann A, Balakrishnan G, Alrubkhi A, Wilke SK, Weber R, Topper B. "Widely tunable lasing in Nd-doped titanate glass prepared by aerodynamic levitation melting." Optics Letters. 2026 Jan 9;51(2):385-8. https://doi.org/10.1364/OL.580940 , Jan-2026
Articles in Peer-reviewed Journals	Weber R, Wilke SK, Rafferty J, Al-Rubkhi A, Benmore C, Moulton B, Kastengren A, Kohara S, Shimonishi R, Koyama C, Ishikawa T. "Structure, properties, and microgravity processing of liquids and glasses." J Ceram Soc Jpn. 2026 May 1;134(5):355-60. https://doi.org/10.2109/jcersj2.26024 , May-2026
Articles in Peer-reviewed Journals	Wilke SK, Wilkinson CJ, Benmore CJ, Neuville DR, Weber R. "Dynamics and structure of molten neodymium titanate." J Am Ceram Soc. 2026 May 26;109(6):e70870. https://doi.org/10.1111/jace.70870 , May-2026